Akashdeep Akashdeep

Experimental Physicist, PhD

Mainz, Germany
+49 157 51084397 | ghalayan.akashdeep@gmail.com
LinkedIn | Google Scholar

Profile snapshot

PhD in Experimental Physics (JGU Mainz, expected Apr 2026) in spintronics and magnetoresistance thin-film devices. Strong in process-to-performance correlation, repeatability, and data-driven workflows (Python).

Skills: Thin-film deposition (sputtering, PLD) | Characterization (XRD, XRR, AFM) | Magnetotransport | Device fabrication (mask design, photolithography, e-beam, etching) | Cleanroom protocols | Design (CAD)

Work Experiences

2021–Present
PhD

Johannes Gutenberg University Mainz, Germany

Research (Spintronics / Altermagnetism)

- Developed **spintronics** thin-film stacks for **magnetoresistance** devices.
- Explored **altermagnetism** in **RuO_2** and **YIG**-based heterostructures.
- Linked **interfaces** and **annealing** to **magnetotransport** performance.

Thin-film process & device engineering

- Deposited stacks by **PLD/sputtering**; controlled interfaces with **vacuum transfer**.
- Integrated **RHEED** into UHV chamber; enabled **in-situ** growth feedback.
- Designed and built a **vacuum suitcase** for contamination-controlled transfers.
- Executed end-to-end **cleanroom** flow: resist, lithography, lift-off, etch.
- Automated cryostat **electrical/magnetotransport** in **LabVIEW**; analyzed in **Python**.
- Characterized films with **XRD/XRR**, **AFM**, **SQUID**; published peer-reviewed work.

2020–2021
Junior Research Fellow

Indian Institute of Technology Delhi, India

- Optimized **Heusler-alloy** sputtering for **repeatable** thin-film growth.
- Correlated **deposition parameters** with **transport** and **magnetic anisotropy**.
- Assembled **sputter hardware**; boosted **uptime** and reliability.

Education

2021–2026	**PhD, Condensed Matter Physics** Johannes Gutenberg University Mainz, Germany
2017–2019	**Master of Science, Physics** Indian Institute of Technology Delhi, India
2014–2017	**Bachelor of Science, Physics** University of Delhi, India

Skills and characterization

Thin-films & device fabrication

- UHV chamber tooling and integration
- DC/RF sputtering, PLD
- Annealing / stack optimization
- Ion etching
- Optical / e-beam lithography

Characterization & measurement

- XRD / XRR
- AFM
- SQUID / FMR / MOKE
- RHEED
- I–V & magnetotransport
- Synchrotron (XMCD, μSR)

Data & design tools

- Python (analysis & plotting)
- SolidWorks
- KLayout
- LaTeX
- Inkscape
- LabVIEW
- Origin

Talks & Conferences (Selected)

- Magnetism and Magnetic Materials / Intermag (Talk) — New Orleans, USA 2025
- International Conference on Magnetic Films and Surfaces (Talk) — Perugia, Italy 2024
- DPG Spring Meeting (Poster) — Dresden, Germany 2023

Awards & Recognition (Selected)

- Featured editorial selections in *Phys. Rev. Appl.* and *Appl. Phys. Lett.* 2025 & 2026
- All India Rank 35 in CSIR-NET (Physical Sciences) — Top 1% nationally 2019
- CBSE Higher Secondary Examination — Top 2% nationally 2014

Research Publications (Full list: Google Scholar)

Surface-localized magnetic order in RuO_2 thin films probed by low-energy muons
Applied Physics Letters (2026) – Thin-film magnetic order characterization.

Interface-driven spin-current magnetoresistance in RuO_2/permalloy heterostructures
Physical Review Applied (2025) – Interface transport quantified for device-relevant validation.

Observation of time-reversal symmetry breaking in the band structure of altermagnetic RuO_2
Science Advances (2024) – Experimental breakthrough revealing a new fundamental class of magnetism.

Languages & Personal Interests

Languages: English (fluent), German (A2–B1), Hindi (native)
Sports: Volleyball, Basketball, Running, Cycling, Swimming, Calisthenics

References

Univ.-Prof. Dr. Mathias Kläui
JGU Mainz — Research Group Head
klaeui@uni-mainz.de

apl. Prof. Dr. Gerhard Jakob
JGU Mainz — PhD Supervisor
jakob@uni-mainz.de